

Mathematics in the Age of Mechanical Reproduction

Statement fidelity, reactivation, and the limits of the verified proof

Christopher Brock

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Abstract. Machine proof abundance separates relations that mathematical institutions have long treated as traveling together: formal correctness, fidelity to the intended problem, human reactivability, access to evaluation, institutional standing, and mathematical significance. Terence Tao’s *Mathematics in the Age of AI* registers the separation of formal correctness from human reactivability and arrives at a criterion: a proof no human can explain is incomplete even when formally verified. I read that as formally complete but epistemically incomplete for publication, uptake, and canonicalization. A 1936 constellation supplies what the criterion needs — Husserl on how a technique outlives the insight it records and on reactivation as the remedy; Benjamin, in an adjacent text, opposing trace to aura, which locates Tao’s “natural friction” and separates it from institutional authority; Turing specifying the machine by abstracting the human computer away, and later bracketing in a footnote the faculty that selects which problems matter at all. Mathematics has since run much of this debate itself, in Jaffe and Quinn’s 1993 proposal for labeling speculative work and Thurston’s 1994 reply. Two mechanisms follow: a statement-fidelity protocol for the unguarded arrow between problem and formalization, and a reactivation packet, with public tests, specifying what must be preserved. Institutional signals are revalidated rather than abolished. Reactivation review at scale remains an open institutional design problem.

0. The problem, and my position

Terence Tao’s ICM essay contains a sentence that ought to stop anyone who works on formalization. His criterion is stated flatly: “A proof that no human can properly explain should be viewed as incomplete” — even where formal verification has succeeded.¹

He reaches it by iteration rather than theory. Solve as many problems as possible — inadequate, since it rewards wrong answers. Solve and verify — inadequate, because a machine can now produce a proof that a kernel accepts and nobody understands, including whoever prompted it. Solve, verify, and communicate — inadequate, because exposition can be over-optimized. Then community acceptance. Then canonicalization. One arrow becomes a chain of five: norms the profession has long practiced but rarely encoded, and never encoded as objectives a machine could be pointed at.²

I think the criterion is right, and that two things are missing from the picture.

¹Terence Tao, “Mathematics in the Age of AI,” arXiv:2608.16753v1 [math.HO], 17 August 2026, 11. This page carries both the expert-talk rule of thumb and the criterion that a proof no human can properly explain should be regarded as incomplete despite formal verification.

²Tao, “Mathematics in the Age of AI,” 5–9. The iterated Goals 6.1–6.5 run across these pages, with the completed pipeline at Figure 4, page 9.

The first sits *before* the diagram begins. Every arrow in Tao’s pipeline is guarded except the first. Verification adjudicates the relation between a formal statement and a proof; nothing adjudicates the relation between that statement and the problem it was meant to express. A subtly weakened statement can be cheaper to prove and still passes every downstream check. I call this the statement-fidelity layer, and §8 proposes a protocol for it — though not, I should say at once, a discovery. Tao names the relation himself, with Tanya Klowden, five months before the ICM essay; what happens to it between the two papers is the subject of §8.³

Six things are being run together in most discussions of this, including at points in Tao’s essay, and the paper needs them apart:

Layer	Relation evaluated	Evidence
Correctness	formal statement $\rightarrow$ proof	kernel, or a published verifier
Statement fidelity	informal problem $\rightarrow$ formal statement	contract, change log, adversarial review
Reactivability	artifact $\rightarrow$ competent reader	packet, reactivation tests
Access	person $\rightarrow$ evaluation procedure	eligibility and tooling rules
Standing	person or result $\rightarrow$ institution	reputation, venue, rank, credential
Significance	result $\rightarrow$ mathematical program	selection, uptake, expert judgment

Formal verification directly addresses the first relation, relative to a stated formal system and trusted base; it does not by itself establish the others. The argument below concerns rows two and three. The remaining relations govern participation, selection, institutional uptake, and the allocation of scarce attention. Keeping the six apart is most of the work.

The second thing missing is an account of what understanding *is* — explicit enough to be built into publication practice, rather than left to practitioner judgment. Mathematics has never lacked the intuition. What it has lacked is a theory with the same institutional standing as its theory of correctness. §§2–7 argue that such an account exists, was independently sought by Gödel, and was independently approximated by Thurston. What 1936 supplies is not the account — the *Logical Investigations* predate it by three decades — but the diagnosis of how a science loses access to one.

One disclosure belongs up front rather than buried. I competed in the SAIR Mathematics Distillation Challenge discussed in §9, submitting two solvers, and I am therefore an interested party in the access arrangements described there. The interest is stated again at that point.

Everything else the argument relies on is on the record: the artifacts, the public competition thread, the cited texts. If the argument is wrong, it should be possible to establish that from those sources alone.

³Klowden and Tao, “Mathematical methods and human thought in the age of AI,” §4.4.

1. The 1936 constellation

Three works converge in 1936, and the convergence is worth a page.

Edmund Husserl published the first parts of *The Crisis of European Sciences and Transcendental Phenomenology* in Belgrade, because Nazi racial policy had made publication in Germany unavailable to him. Walter Benjamin's artwork essay appeared in the *Zeitschrift für Sozialforschung*, issued from the Institute's exile in Paris. And in Cambridge, Alan Turing's "On Computable Numbers" was received on 28 May 1936 and read that November; it appeared in print in 1937, so what converged in 1936 was the work rather than the publication.

Two of the three concern *Technik*. Benjamin's title turns on *technische Reproduzierbarkeit*, technological reproducibility; Husserl's diagnosis in §9 of the *Crisis* is *Technisierung*. Both ask what becomes of a human achievement once the means of producing it are good enough to run without it. Turing specified such a means.

I offer the year as juxtaposition, not prophecy. None of the three anticipated machine mathematics and nothing here depends on suggesting they did. They analyzed a structure; the structure recurs; their analyses are further along than anything the present discussion has assembled from scratch.

The circumstances under which Husserl wrote, and the fate of his manuscripts and his assistants, bear on the argument and are set out in §12, where they can be read against a case rather than as an overture.

2. What Husserl saw

The *Crisis* is not about paradoxes in set theory. It is about what happens to a science when its methods become powerful, reliable, and easy to transmit.

The central case is Galileo. Husserl's claim is that mathematical physics performed a substitution — replacing the world as encountered with a mathematical manifold that models it — and then forgot that a substitution had occurred. The model is an *Ideenkleid*, a garb of ideas: we use it to disclose the world and eventually take it for the world.⁴

The mechanism is technization. A method begins as an achievement of insight; someone saw something and built a procedure preserving what was seen. The procedure works, is taught, is refined, is scaled. Eventually it becomes possible, then normal, to operate it correctly without recovering the insight it came from. Husserl's term for what happens to the meaning is *Sinnentleerung*: the sense drains while the technique continues to function perfectly. The continued success is what makes the loss difficult to see.

This is not an argument against technique. Technization is a condition of scale: nobody reconstructs the calculus before taking a derivative, and a discipline demanding it would produce almost nothing. The narrower claim is that a technical tradition lives from deposits of achieved insight, and that a discipline losing the capacity to reactivate those deposits may retain every formula while losing access to what made them knowledge.

⁴Husserl, *Crisis*, §9, where the Galileo analysis, the mathematization of nature, and the *Ideenkleid* are developed; section numbers are stable across editions. For a scholarly reconstruction see J. N. Mohanty, *Edmund Husserl's Freiburg Years, 1916–1938* (New Haven: Yale University Press, 2011), 394–400, esp. 398–99, on idealization, arithmetization, technical thinking, and Galileo as at once discovery and concealment.

The response is in the same volume, in the text known as *The Origin of Geometry*. Achieved meaning is written down, and writing **sediments** it — making the achievement stable and transmissible, and simultaneously permitting it to be handled by someone who never recovered its originating act. What keeps a tradition alive is **reactivation**: a person recovering, from the deposit, the sense-making that the deposit records.⁵

3. What Benjamin saw

Benjamin's essay concerns what happens to a work of art once it can be reproduced at will.

His term for what reproduction destroys is **aura**: the work's unique presence in time and space, its embeddedness in tradition, the authority of having been in one place with a history. A reproduction does not carry the original's here-and-now or its accumulated historical testimony; it brings the object into the recipient's situation and thereby contracts auratic distance.⁶

Benjamin sharpened the term elsewhere, and the sharpening is what this paper needs. In the *Arcades Project* he sets **trace** against aura directly. The trace is the appearance of a nearness, however far off whatever left it may be; the aura is the appearance of a distance, however close the thing that summons it. His conclusion is the operative one: in the trace we take possession of the thing, and in the aura the thing takes possession of us.⁷

Trace and aura are therefore not two components of one phenomenon. They are opposed. That will matter in §6 and again in §10, and I flag it here because the looser reading — aura as an undifferentiated atmosphere containing both — is the one a reader is likely to arrive with.

The distinction that matters for this paper is between **cult value** and **exhibition value**. In its origins the artwork served ritual: what mattered was that it existed, in the right place, tended by the right people; being seen was secondary. Cult value attaches to position, and position is administered. Exhibition value concerns public presentability: reproduction vastly expands the situations and audiences in which a work can be encountered.⁸

Benjamin does not simply mourn the aura. He argues that mechanical reproduction emancipates the work from ritual and transforms its social function, and his examples show cult value organized materially through restricted visibility.⁹ The further claim — that this weakens custodial authority in a way transferable to mathematical institutions — is the extension I draw from his analysis, not a thesis he advances.

⁵Husserl, "The Origin of Geometry," appendix to the *Crisis*. Mohanty, *Freiburg Years*, 420–24, esp. 421–23, emphasizes the double movement this paper turns on: writing gives ideal meaning an enduring intersubjective existence and makes reactivation necessary once the originating evidence has disappeared.

⁶Walter Benjamin, "The Work of Art in the Age of Mechanical Reproduction," in *Illuminations*, trans. Harry Zohn, ed. Hannah Arendt (New York: Schocken Books, 2007), §§II–III, 220–23. Benjamin identifies the spatial distance of aura with cult value, and the cult image with unapproachability, at 243 n. 5 — which grounds the aura/cult link used in §10, though not the further application to academic standing, which remains mine.

⁷Benjamin, *The Arcades Project*, Convoluted M, [M16a,4]. The convoluted number is the stable locator across editions.

⁸Benjamin, "Work of Art," §§IV–V, 223–25. His examples show cult value materially organized through restricted visibility — images accessible only to priests, kept covered for most of the year, or fixed beyond ordinary view (224–25).

⁹Benjamin, "Work of Art," 221 and 224: the former on reproduction detaching the object from tradition, the latter on emancipation from ritual dependence.

Two texts, then, with opposite valences. Husserl: technique scales and meaning drains, which is a loss. Benjamin: technique scales and authority collapses, which is a gain. Mathematics under machine proof is exposed to both, and the rest of this paper is about that tension.

The pairing is not mine to claim either. Klowden and Tao open their historical section by asking whether an original artwork has aesthetic or creative features that no mechanical reproduction could properly capture, and then name what modern AI has produced: an unprecedented decoupling between the outward form of an intellectual product and the values and thought processes that made it. That is the form/process decoupling this paper's six-relation table tries to make precise, and it is why the title is a recovery rather than a conceit.¹⁰

4. What Turing built

The third text is not a diagnosis. It is a construction, and what deserves attention is how it proceeds.

Turing does not begin with hardware or with an axiom system. He begins with a person. Throughout the paper "computer" denotes a human being doing arithmetic. In its ninth section he observes that such a person's behavior at any moment is determined by the symbols currently observed and by what he calls the state of mind; he then argues the number of distinguishable states of mind must be finite, since infinitely many would include states arbitrarily close together and liable to confusion. Having bounded the observed symbols and the states of mind, he specifies an abstract model of effective computation reproducing exactly the operations that remain.¹¹

Set that sequence beside §2. Begin with an act performed by a subject. Describe it as procedure. Bound it. Set the subject aside. Retain the procedure, which now runs without anyone.

The strong form of this comparison is wrong and I do not want it. Turing was not draining sense from a tradition; he was building a definition, and one of the century's great ones. Abstraction and *Sinnentleerung* are different things.

They are the same shape, though: each keeps the operable residue of an act and sets aside whatever made it an act. Turing did this deliberately and to enormous profit, and the setting-aside is what makes the model possible. Nothing here is a criticism of him. It is a description of what got left on the desk, and it stays uninteresting until the model's output comes back to people who have to make sense of it.

Ninety years later, descendants of that model are writing proofs. When Tao asks what to do about a verified proof no human understands, he is asking what happens when the residue is returned at volume, to a community that must reactivate it, from a construction that had no provision for putting the subject back.

¹⁰Klowden and Tao, "Mathematical methods and human thought," §§2.1–2.2.

¹¹A. M. Turing, "On Computable Numbers, with an Application to the Entscheidungsproblem," in *The Essential Turing*, ed. B. Jack Copeland (Oxford: Clarendon Press, 2004), 75–77, esp. 75. Turing begins from the human computer, the symbols under observation, and the computer's "state of mind;" the finiteness argument and construction continue through 77.

Turing's own limit

There is a second point, and it answers the strongest objection to this paper — that mathematicians already know this and need no phenomenology to see it.

They do already know it, and Turing said so in the discipline's own idiom three years later.

His 1939 paper on systems of logic based on ordinals — the Princeton thesis written under Church — asks how far the incompleteness limitation can be pushed back by iterating formal extensions. In its course he distinguishes two faculties in mathematical reasoning: **intuition**, consisting in spontaneous judgments that are not the outcome of conscious trains of reasoning, and **ingenuity**, the work of arranging matters so that fewer such judgments are required. His conclusion is this paper's thesis reached from the other direction: formalization can reduce the burden borne by intuition and cannot remove it.¹²

The same paper introduces the **oracle machine**, and the tempting reading of it is wrong. Turing's oracle is a technical device for studying relative computability and the ordinal-indexed hierarchy of logics. It is not a name for mathematical intuition.¹³

The paper establishes two things: that a formal system's reach can be extended indefinitely without becoming self-sufficient, and, separately and in prose, that intuition stays in play. Two results. Fusing them overstates the case, and the case does not need it.

One more thing in that paper, and it may be the most contemporary sentence Turing wrote. The passage introducing intuition and ingenuity carries a footnote in which he sets aside what he calls "that most important faculty which distinguishes topics of interest from others", adding that for present purposes he treats the mathematician's function as simply determining whether propositions are true or false.

The most important faculty, bracketed in a footnote, so that the argument can proceed.¹⁴

That was the correct move in 1939, and it is the move Tao's pipeline inherits. Every stage from generation to canonicalization presupposes that someone has already decided the problem was worth solving. Under proof scarcity, selection and proof production stayed coupled in practice: choosing a problem generally committed scarce human effort to solving it, so the choosing and the doing were the same decision. Machine abundance breaks that coupling. When derivations are cheap, deciding what deserves scarce human reactivation becomes part of the mathematical work rather than a precondition of it. That is the sixth row of the table in §0, and no protocol in this paper addresses it.

Nor is the coupling argument mine. Klowden and Tao put it in a footnote of their own: in earlier eras, when a rigorous proof required painstaking human effort, mathematical activity naturally concentrated on problems the community judged interesting — a largely social mechanism that could correct itself because the literature moved slowly enough. Where results can be mass-produced, they write, such questions may require far more active attention.¹⁵ Turing

¹²Turing, "Systems of Logic Based on Ordinals," in *The Essential Turing*, §11, 192–93. The intuition/ingenuity distinction and n. 25 are on 192; the conclusion that no formal logic wholly eliminates the necessity of intuition is on 193.

¹³Turing, "Systems of Logic Based on Ordinals," §4, 156–57, where the oracle and the o-machine are introduced.

¹⁴A. M. Turing, "Systems of Logic Based on Ordinals," §11 and n. 25, in *The Essential Turing*, ed. B. Jack Copeland (Oxford: Clarendon Press, 2004), 192.

¹⁵Klowden and Tao, "Mathematical methods and human thought," n. 17; see also their nn. 11 and 12, on the need for standard procedures for citing unverified machine "guesses," and on the literature's habit of publishing successful proofs while discarding the processes that produced them. Nor am I alone in leaving it open: the Leiden Declaration names the same danger from the other direction, warning that research questions may come to be prioritized by

bracketed the faculty in 1939 so that his argument could proceed. Tao unbrackets it in 2026 and finds it waiting.

So the claim is not that mathematics needs phenomenology to notice that verification and understanding differ. It is that mathematics established the formal limit rigorously but never institutionalized a comparably explicit account of what occupies the space that limit leaves. Turing marked the place. Husserl described what stands there, how it functions, and how it can be lost.

5. The subject, and why it does not make mathematics relative

Mathematical institutions absorbed Turing’s framework far more directly than either of the other two. The reason is a fork that occurred earlier.

Frege and Husserl begin close together: both destroy psychologism, and neither allows mathematical truth to be a fact about what happens in heads. They part over what follows. Frege holds that thoughts subsist objectively and that philosophy’s task is to describe their logical structure; minds *grasp* those thoughts, and grasping is left unexplained and consigned to psychology. The linguistic turn follows, and the question of the act recedes. Husserl holds that ideal meanings are real *and* constituted in acts describable without reduction to psychology — fighting on two fronts, against psychologism and against a formalism with no subject at all.

The consequence needs stating carefully, since the reductive version is easy to refute. Mathematics did not end up with *no* account of understanding. Working mathematicians have always had a rich informal grasp of what understanding a proof involves, and Thurston’s 1994 essay — which Tao quotes — is a formidable account written with no phenomenological apparatus whatever. What the discipline lacked was an account with the same **institutional standing** as its account of correctness: something explicit enough to be built into verification and publication practice.

The difference is structure rather than prestige, and it is worth being exact about, because the obvious reply is that Thurston reached these phenomena without any phenomenology and is easier to read. He did, and he is. But what he offers is a taxonomy — seven ways of conceiving the derivative, an inventory of mental faculties — and a taxonomy cannot be built into a rule. It tells an editor that understanding is various; it does not tell her what a submission must contain. Husserl offers something narrower and more useful for that purpose: a distinction between *kinds of act*, from which it follows that no accumulation of verification becomes understanding, because the two are not quantities of the same act at all. That is a claim a publication standard can be built on, since it specifies what the standard has to test for. I return to this in §7. Understanding remained practitioner judgment while correctness became a matter of specifiable rules. The asymmetry cost little while understanding and proof traveled together; proof assistants and large formal libraries had already begun to strain it, and machine generation removes the last conditions under which it could be ignored.

One major figure pursued the gap. From 1959 until his death in 1978 Gödel read Husserl closely and continuously, and in a lecture prepared but never published argued that phenomenology offered the right framework for grounding mathematical intuition.¹⁶ I want to be careful

their amenability to automation rather than by expert judgment of their significance, and that broader understanding of a field may be permanently lost in the process.[^{leiden-significance}]

¹⁶Mark van Atten and Juliette Kennedy, “On the Philosophical Development of Kurt Gödel,” *Bulletin of Symbolic Logic* 9, no. 4 (2003): 425–476, esp. 425ff. Gödel’s own lecture is “The Modern Development of the Foundations

about the route: incompleteness forms part of the background, and it is tempting to say the theorem drove him to Husserl, but his path ran through Leibniz and his own Platonism as well, and the historical picture is more tangled than that. What is not in doubt is the destination. The figure whose theorem opens Tao's prologue spent the last two decades of his life looking for an account of mathematical intuition, and looked for it in Husserl.

Now the objection that must be answered before anything else: does restoring the subject make mathematics relative?

It does not. The subject Husserl restores is **transcendental, not empirical** — not this person on this afternoon, but the invariant structure of any consciousness for which a mathematical object could be given. And it **constitutes**, which is not to **construct**. Constitution is the achievement by which something *identical* shows itself across indefinitely many acts, times, and persons. The identity is the point: the theorem I grasp and the theorem you grasp are numerically the same object, and they are the same because constitution follows an invariant structure rather than a personal one.¹⁷

Objectivity is not surrendered; it is explained. Frege leaves the subject's relation to objective thought comparatively undertheorized for the purpose at issue here — not a defect in his project, and a gap for ours.¹⁸

The error to avoid is easy to state as a rule. Any account on which a truth becomes *actual* when a mind attends to it has returned to psychologism. The subject is not what makes a theorem true; it is what makes a theorem **available**. Formal correctness is established by proof under the relevant rules. Availability concerns whether the proof's mathematical sense can be recovered by anyone. Everything below concerns availability.

One precision, since the loose version invites a fair objection: the machines have not made truth cheap. What has become cheap is the *production of formally checkable artifacts* — which is exactly what pulls correctness apart from availability, since artifacts now accumulate faster than the acts that would make their sense recoverable.

6. Tao's pipeline, and friction

Return to the chain of five from §0. Read through the *Sixth Logical Investigation*, it stops looking like process management.

Husserl distinguishes an intention that is merely **signitive** from one that is **intuitively fulfilled**. A signitive intention points toward its object through signs without giving it; in fulfillment, what was emptily intended becomes present in a corresponding intuition. *Evidenz*, technically, is not a feeling of confidence but the experience of the intended matter as given. The vocabulary is

of Mathematics in the Light of Philosophy," in *Collected Works*, vol. III, ed. Solomon Feferman et al. (Oxford: Oxford University Press, 1995), 374–387.

¹⁷Edmund Husserl, "The Way into Phenomenological Transcendental Philosophy by Inquiring Back from the Pre-given Life-World," in *The Phenomenology Reader*, ed. Dermot Moran and Timothy Mooney (London: Routledge, 2002), 151–74, esp. 154–60, on ideal objectivity across subjects, the shared world, and science as a human accomplishment; Mohanty, *Freiburg Years*, 421–22.

¹⁸How far Frege's 1894 review shifted Husserl's position is disputed: Føllesdal argues for substantial influence, Mohanty that Husserl had already moved. I take no position.

Husserl’s own and load-bearing in his hands: the task is to trace knowledge back to “an adequate fulfillment in intuition”, phenomenology having no business resting in mere words.¹⁹

A machine-checked proof is not necessarily devoid of intuitive content — a mathematician may read it, reconstruct it, and recover what it records. But for a reader who cannot perform that reconstruction, the reader’s relation to the accepted artifact remains largely signitive: the certificate indicates a valid derivation exists without making the state of affairs evident to that reader. Formal adequacy can be complete while fulfillment remains absent for that reader. Verification does not become understanding by accumulating, because they are not quantities of the same act.

That is why the iteration could not terminate at Goal 6.2. It also explains why the chain runs through more than one relation rather than five instances of one. Exposition and communication bear on fulfillment. Acceptance and canonicalization concern uptake, standing, and sedimentation into a tradition — different rows of the §0 table, which is why no single mechanism serves all of them.

Benjamin explains the essay’s most striking paragraph, which Tao leaves undeveloped. In a human proof, Tao observes, the passages the author found difficult retain **natural friction** — an apologetic remark, an over-careful lemma, a paragraph visibly rewritten — which tells the reader where to slow down and is a principal channel for a field’s tacit knowledge. He illustrates it with a photograph of a 1991 Bourgain paper covered in the annotations of his younger and very frustrated self, and notes that fighting through such texts is how he came to understand Bourgain’s way of thinking.²⁰

In Benjamin’s vocabulary, friction is **trace** rather than aura, and the distinction does real work. The apologetic remark, the over-careful lemma, the paragraph visibly rewritten, the annotated page — each brings a vanished act of struggle *near*, which is exactly what Benjamin says a trace does. Aura would do the opposite: hold the work at a distance even while it stands open in front of you. Over-polished exposition sands the trace away along with the typos, and Tao’s own formulation is the sharp one: the result is a text “easy to read and hard to learn from.”²¹

The observation has a precursor in his own work, in a vocabulary worth keeping. In March he and Klowden describe the “smell” that experienced mathematicians use to judge an argument before checking it line by line, and warn that a system trained on formal correctness alone could produce **odorless** proofs: superficially resembling well-written human work, passing verification, and yielding far less insight than expected. They ask directly whether something is lost, in a world of uniformly AI-polished output, in forsaking the grubbier and messier world of hand-typed text. And they supply the specimen this paper otherwise lacks — the AlphaProof solutions to the 2024 IMO problems, formally verified and containing numerous redundant or inexplicable steps.²²

¹⁹Husserl, *Logical Investigations*, Sixth Investigation §§1–16 (intention and fulfillment) and §§36–39 (evidence). The phrases quoted are from Husserl’s “Introduction to the *Logical Investigations*,” in *The Phenomenology Reader*, 65–77, at 76 and 67 respectively; page references for quotations are to the *Reader*, whose text is the 1970 Findlay translation, and not to the 2001 edition.

²⁰Tao, “Mathematics in the Age of AI,” 7, including Figure 3 — natural friction, tacit knowledge, over-polished exposition, and the annotated Bourgain page.

²¹Tao, “Mathematics in the Age of AI,” 7, including Figure 3 — natural friction, tacit knowledge, over-polished exposition, and the annotated Bourgain page.

²²Klowden and Tao, “Mathematical methods and human thought,” §§4.2 and 4.4, and n. 16. “Odorless” is their term; the sensory metaphor is borrowed, they note, from the notion of a “code smell” in software engineering.

Two consequences follow, and §10 turns on both.

First, the friction Tao values is not auratic and never was. It is the mark of the hand, and it aids reactivation because nearness is what it produces.

Second, and less obviously, institutional prestige can operate *auratically* on a mathematical result. A famous theorem in a famous journal is entirely visible and still held at a distance: available to read but practically costly to contest, because the authority to judge it sits elsewhere. On the institutional extension proposed here, that has an auratic structure: the appearance of a distance, however close the thing stands. Benjamin’s own gloss makes the extension testable rather than merely evocative — “The essentially distant object is the unapproachable one” — which converts “cult” from a synonym for prestige into a specific relation of practical unapproachability.²³

7. The last time mathematics had this argument

Mathematics has run much of this debate itself, in its own literature.

In July 1993 the *Bulletin of the AMS* published Jaffe and Quinn’s “Theoretical Mathematics,” occasioned by speculative work flowing in from theoretical physics and worried about a bifurcation mathematics had never had, between those who conjecture and those who verify. Of the four failure modes they name, the third matters here: when a theorizer claims full credit for a result they have not established, the rigorous work needed to make it reliable becomes unrewarding, nobody volunteers, and the area dies — not from exhaustion but from debris. They called it a **dead area**.²⁴

Set that beside Tao’s report that the Erdős problems database already holds dozens of AI-generated submissions, many probably correct, which no expert has volunteered to verify — several submitted by people stating outright that they are not qualified to.²⁵

Same mechanism, machine scale, with a claimant who cannot be argued with about credit.

Jaffe and Quinn’s remedy was nomenclature: in theoretical material *conjecture* should replace *theorem*, *predict* should replace *show*, *supporting argument* should replace *proof*, and titles should carry a flag word so status travels with the citation. They also proposed reserving a major share of credit for the work that validates a result. Tao’s call to shift emphasis from proof generation toward digestion is that credit argument restated; the Leiden Declaration’s insistence that credit and responsibility remain with humans is the same clause under a condition Jaffe and Quinn never faced — a speculator with whom credit cannot be shared because it cannot receive any.²⁶

Their nomenclature proposal anticipates the epistemic-state badges of §10, and I claim no originality for the principle — only that it extends to computational exploration, that status

²³Benjamin, “Work of Art,” 243 n. 5.

²⁴Arthur Jaffe and Frank Quinn, “‘Theoretical Mathematics’: Toward a Cultural Synthesis of Mathematics and Theoretical Physics,” *Bulletin of the American Mathematical Society* 29, no. 1 (1993): 1–13, at 8–9; arXiv:math/9307227v1. The four-part list under “The Problems,” including the dead-area mechanism, is on 8; the discussion of credit and the disincentive to perform rigorous cleanup is on 9.

²⁵Tao, “Mathematics in the Age of AI,” 6.

²⁶*Leiden Declaration*, 5, carrying “Support the needs of reviewing,” “Retain the responsibility for correctness,” “Affirm the humanity of authorship,” and “Put effort into proper attribution.” Note that the responsibility clause assigns human authors responsibility for the *adequacy* of arguments and results, not only their correctness — which is the statement-fidelity layer of §8 under another name.

marking and automated consistency checking are now mechanically cheap where in 1993 the whole burden fell on editorial will, and that machine output has made its absence expensive. Their own name for the whole package was “truth in advertising,” which remains the right one.²⁷

Thurston’s reply

The following April Thurston answered them, and his essay is the one Tao quotes.

His objection is that Jaffe and Quinn projected the sociology of mathematics onto a single axis, speculation versus rigor, which omits the phenomena. He proposes a different question: how mathematicians advance **human understanding of mathematics**. And he claims that reliability does not primarily come from formal checking of formal arguments but from mathematicians thinking carefully and critically, with knowledge living in the minds and social fabric of a community that documents support but does not constitute.

Then the case. Entering foliation theory as a graduate student, he proved a series of major results quickly, and within a few years the field emptied; people were advising students away from it. His retrospective names two contributing factors, both his own: he had written the results in a formidable style leaning on background he possessed and readers did not, which raised the entry barrier and discouraged people from learning the proofs of key theorems; and little credit was left in the area for anyone else.

Not about formal verification, machines, or 2026 — but the mechanism is the one tracked here, and documented in the *Bulletin* by the person best placed to describe it: correct results, correctly recorded, whose sense did not survive the recording in a form others could recover.

What he did next matters more. Working on three-manifolds and the geometrization conjecture he changed method deliberately: wrote notes and mailed them, eventually to some twelve hundred people; concentrated on infrastructure — definitions, ways of thinking, mental models — and was slow to publish proofs of theorems he could prove, so that others had room to discover and take credit. The field thrived, and other proofs came from other people.

That is reactivation adopted as working practice by someone who had watched the alternative once.²⁸

Two observations there do further work. Thurston describes a **huge expansion factor** between how an idea sits in his own thinking and what must be said to convey it; I read that, interpretively, as the signitive-to-intuitive distance reported from the inside, though his essay establishes the phenomenon and not the reading. And he notes that proofs depend on the audience: a step conveyable to topologists in two minutes took an hour with analysts. Reactivation is not a property of a text but a relation between text and reader.

Where Thurston cuts against me

Thurston reports that mathematicians accepted his geometrization result quickly — on the available documentation, their experience of him, and the opinions of experts he had spent time with.

²⁷Jaffe and Quinn, “‘Theoretical Mathematics,’” 10–11. Their prescriptions reserve substantial credit for rigorous validation and recommend replacing “theorem,” “show,” and “proof” with status-marking terms such as “conjecture,” “predict,” and “supporting argument,” with theoretical status flagged in titles and abstracts.

²⁸Thurston, “On Proof and Progress in Mathematics,” §§1–6; the foliations episode, the change of method on geometrization, the expansion factor, and the audience-dependence of proof are all in §6, and the understanding question is posed at the head of §2.

That is standing doing epistemic work, and doing it well: efficient, approximately correct, and cheaper than artifact-based verification would have been.

So the cleanest case of positional authority functioning as a **reliable signal** is in this literature too. My claim is not that the mechanism was illegitimate but that it was difficult to audit independently — someone outside that circle had no practical way to check the judgment and no standing to contest it — and that its reliability declines as capability, understanding, and position cease to correlate.

Klowden and Tao describe the mechanism itself, borrowing a metaphor from Kim: a professional mathematician accumulates credibility as a kind of currency, by proving difficult new results, and only then can *afford to spend* it on speculative activity — formulating conjectures, or philosophizing about what a result means.²⁹ That is an unusually candid account of how standing actually works, and it has a consequence the metaphor makes plain. Under scarcity the currency was earned in the same coin it was spent: you proved things, then you were listened to. Machine abundance devalues the earning side without touching the spending side, so the question of who is heard on the goals of mathematics is decided increasingly by a balance accumulated under the old rate.

The opening, and a caution about it

Having rejected the speculation-versus-rigor axis, Thurston asks how people understand mathematics, says the question is very hard, that understanding is individual and internal and difficult even to be aware of, and that he can only touch on it lightly. He then offers a practitioner's partial account — seven ways of conceiving the derivative, a taxonomy of mental facilities — and moves on.

I read that as an opening. The skeptical reading deserves stating: Thurston may have been right that the question resists the kind of explicit general theory Husserl attempted, and that understanding's multi-modal, audience-dependent character does not yield invariant structure.

The practical proposals below do not require assent to Husserl's complete transcendental project. But one thing in it stays load-bearing under any version of that skepticism: the act-type distinction, which is what explains why additional verification, considered strictly as verification, cannot entail reactivation by a reader. Drop that and the criterion becomes a preference about how mathematicians ought to write. Keep it and the criterion has a reason.

8. The unguarded arrow

Now the gap named in §0, in full.

Tao's pipeline opens with *open problems* → *unverified solutions*, and every arrow downstream is guarded: verification for correctness, exposition for legibility, refereeing for acceptance, canonicalization for integration. The first arrow is unguarded. Formal verification adjudicates the relation between a **statement** and a **proof**; it is silent on the relation between the statement and the problem the statement was meant to express.

This is the most Goodhart-exposed point in the structure, because a subtly weakened statement can be *cheaper to prove* and still passes every downstream check. A quantifier reordered,

²⁹Klowden and Tao, "Mathematical methods and human thought," n. 15, citing Minhyong Kim.

a hypothesis strengthened, an object defined so the hard instances fall outside it — each yields an artifact that verifies, reads cleanly, and answers a question nobody asked.

Husserl’s account of **categorial intuition** does not amount to a theory of specification fidelity, and I do not want to claim that it does. What it explains is one indispensable feature of the informal side: grasping a state of affairs — *that* this implies that — as given, rather than merely manipulating its expression. A state of affairs is a structured unity, not an object, and something has to give it before anything can answer to it.³⁰

The layer is not my discovery, and the record matters. Five months before the ICM essay, Tao stated it outright with Klowden: formal verification certifies that a formalized argument establishes a formal statement, and does not rule out errors of translation between that statement and the one originally intended. Their example is better than any abstract formulation. Fermat’s Last Theorem asserts that no natural numbers a, b, c satisfy $a^n + b^n = c^n$ for $n > 2$; implicit in the informal statement is that the naturals begin at 1. An AI permitted to let a, b, c be zero can produce a formally certified proof that Fermat’s Last Theorem is *false*. The same paper adds, in a footnote, that mathematics could in principle be “hacked” by subtly manipulating definitions in a standard library such as Mathlib — which is the trusted-base concern of §10 and the cheaper-to-prove concern of this section, from the same source.³¹

What is worth remarking is what became of it. Stated plainly in March, the relation has by the August pipeline dropped out of the diagram entirely: every arrow guarded, verification standing at the head of the chain, and no node for the translation that precedes it. I do not think this was carelessness. I think it is what happens to a relation that has no stage, no name, and no one whose job it is — which is the argument for giving it all three. The protocol below is a response to that disappearance, not to any doubt that the gap exists.

This is not a philosopher’s distinction imposed from outside. Lean’s own validation guidance draws it in the same terms, stating that it matters to distinguish whether a theorem has a valid proof from what the theorem statement means, and then presenting an escalating sequence of checks that audit derivability, dependencies, and implementation trust while leaving that second question open. Its list of remaining assumptions ends where this section begins: no human error or misleading presentation of the statement in the trusted challenge file, and if there is doubt that a theorem means what it appears to mean, the statement and every referenced definition must be investigated by someone.³²

The formulation to avoid is that formalization simply cannot do this. Recent systems already operationalize parts of the layer: ProofBridge evaluates generated formalizations against reference formalizations through kernel-checked bidirectional implication, and FormalMATH combines several autoformalizers with back-translation-based semantic agreement, negation-based

³⁰Husserl, *Logical Investigations*, Sixth Investigation §§40–48. For a compact secondary statement of the point at issue — the difference between grasping that $2+2=4$ and manipulating the symbols correctly — see Dermot Moran, “Editor’s Introduction,” in *The Phenomenology Reader*, 8–9.

³¹Tanya Klowden and Terence Tao, “Mathematical methods and human thought in the age of AI,” arXiv:2603.26524v1 [math.HO], 27 March 2026, §4.4 and n. 7. The paper is an unabridged version of a solicited article for the *Blackwell Companion to the Philosophy of Mathematics*.

³²Lean Project, “Validating a Lean Proof,” *Lean Language Reference*, accessed 20 August 2026, <https://lean-lang.org/doc/reference/latest/ValidatingProofs/>.

disproof filtering, and final expert review.³³ These substantially reduce fidelity risk. Each of them also requires a second representation or an interpretive judgment, and none follows from proof verification alone.

The defensible claim is therefore narrow and survives every improvement in tooling. **No procedure operating solely on a target proposition and its proof can establish fidelity to a prior informal intention.** The reason is underdetermination: the same proposition–proof pair is compatible with indefinitely many prior intentions, including none at all. The relation to be certified is simply absent from the procedure’s inputs.

Diagnose a gap, invoke categorial intuition, and you have offered philosophy where engineering was wanted. Under abundance, “more expert judgment” is the scarcest remedy anyone could name. The division of labor has to be stated, then. Phenomenology explains why proof verification alone cannot eliminate the layer. Staffing it is a design problem, and design problems take partial solutions:

1. **Formalization contract.** Publish the informal problem and the formal theorem adjacent, as a required artifact rather than commentary.
2. **Semantic change log.** Enumerate every rendering decision: quantifier order, hypothesis strength, domain restrictions, definitional choices, excluded cases. Most fidelity failures are invisible because these are never written down.
3. **Adversarial statement review.** Assign a reviewer whose task is not to check the proof but to hunt for weakenings — what the statement fails to cover that the problem intended. A different job from refereeing, and a narrower one.
4. **Dual formalization.** For load-bearing results, require two independently produced formal statements, or translation into a second system. Independence has to be real: two renderings from one model on one prompt are one rendering. Vary the authoring process, the model family, or the semantic representation. Divergence localizes the question.
5. **Boundary testing.** Generate instances just inside and outside each hypothesis; accidental restriction shows up as cases the statement silently declines to address.

One failure class the protocol does not catch deserves naming, because I ran into it in my own work. A formalization can be honest, verified, and empty of content: a result conditioning the Riemann Hypothesis on a hypothesis provably equivalent to it is a theorem, correctly proved, that establishes nothing. Nothing in the five steps above detects this, since the statement is faithful to a real problem and the proof genuinely establishes it. Under abundance, a flood of machine-generated reductions will contain many such tautological loops, and distinguishing a genuine reduction from a circular repackaging becomes a first-class problem — plausibly an automatable one, and certainly not one this paper solves.

None of this closes the gap by fiat. Formal specification, semantic modelling, independent translation, and adversarial testing reduce it substantially without eliminating it. The defensible statement is narrow: **formal verification cannot, by itself, certify that a formal statement remains faithful to the preformal problem it was intended to express.** That is what makes the layer necessary. The protocol makes the judgment that remains explicit and more tractable.

³³Prithwish Jana et al., “ProofBridge: Auto-Formalization of Natural Language Proofs in Lean via Joint Embeddings,” arXiv:2510.15681; Zhouliang Yu et al., “FormalMATH: Benchmarking Formal Mathematical Reasoning of Large Language Models,” arXiv:2505.02735, §3.2.

9. One case, disclosed

A single episode, disclosed. It follows the Thurston material because that case is stronger and needs none of my testimony. This one shows something Thurston’s does not: what happens when access to the *criterion* is unevenly distributed, and what happens when that is put right.

Between March and April 2026 the SAIR Foundation ran the Mathematics Distillation Challenge on equational theories, an outgrowth of Tao’s Equational Theories Project, discussed on an open Zulip stream with Tao participating. I competed, and I am an interested party in the access arrangements below — which is both why I noticed them and a reason to discount my reading of them. This is interested testimony about one case, not an audit.

The correctness criterion was published. For Stage 2, where each answer carries a machine-verifiable Lean certificate, SAIR released the verifier itself under an open license: the Lean modules, the evaluation pipeline, the adversarial test manifests, a local harness, and public practice sets. The repository states that the same judge code runs locally and at official evaluation, so a solver turning the harness green locally should receive the same verdict in production; environments can still differ, and the claim is theirs rather than something I have independently confirmed. Anyone at all can now establish that their certificates are accepted by the verifier the organizers use, and the anti-Goodhart reasoning survives intact, because final evaluation still runs against a held-back set the repository deliberately omits.

That settlement was not obvious in advance. Earlier in the competition the evaluation format was kept opaque, on Tao’s stated reasoning that publishing it would invite overfitting rather than distillation of insight that generalizes — a decision I think was also right. Opacity’s cost is distributional rather than epistemic: teams able to reconstruct the apparatus locally did so, and the constraint bound only on those who could not.³⁴

Access to the standing-conferring evaluation opened later, and quietly. Reproducing the verdict function is not being evaluated. Eligibility is not itself standing, though official evaluation is what confers it here. Access to official assessment against the held-back set was for months conditioned on prior-stage participation; I requested it in May and was told the restriction was intended. On 20 August 2026, while completing this draft, I checked again and found registration available. I cannot date the change, and I learned of it by looking rather than by any notice reaching me. *(The restriction, the May exchange, and the discovery are my own account. The repository, its license, and the held-back-set design are on the public record.)*

Three lessons. The available record supports an explanation by omission, and I attribute no exclusionary motive.

Opaque criteria distribute verification capacity unevenly. Publishing the criterion makes the correctness criterion publicly reproducible, which SAIR did without regard to institutional position. And third, the one I would not have seen except as an affected participant: **a policy change that is not dated and announced does not reach the people it releases.** It reaches whoever happens to look again. For some period I could have entered work for official assessment and

³⁴The stream record includes repeated requests for the parser and decoding parameters, a team announcement of a locally reconstructed harness, an analysis showing the platform’s credit-*hold* mechanism rather than actual cost bound testing capacity, and a mid-competition change to the submission format announced in-stream. Dated archival copies of each message are retained by the author. No permalinks are reproduced here; readers who wish to verify these particular claims should treat them as weaker than the repository and competition-page evidence cited above, which is directly addressable.

did not know — not because access remained closed, but because I found no announcement in the channels I reviewed and no notice reached me.

That communication gap is the cheapest of the three to close, and it is what §10's access recommendation addresses. Knowing that the rules have changed is itself a form of access, and it is the form most easily overlooked, because from inside an institution a change feels complete the moment it ships.

10. Destroy the cult, keep the trace

Tao's rule of thumb: if authors cannot convincingly demonstrate they can give a clear, expert-level talk on their result, correctly and with proper attribution, it should not be published.

The rule does two jobs, and only one is epistemic. As an epistemic standard it asks whether reactivation can be demonstrated under a public standard — an outcome that may depend on the evaluator's competence, and must not depend on the evaluator's institutional identity. As an operational standard it determines who receives the opportunity to demonstrate that capacity at all, and that is settled by whoever decides who stands in the room. Under scarcity the two readings were nearly indistinguishable. Abundance splits them, and every rule phrased as *the community must accept it* quietly becomes a rule about who gets to try.

Institutional authority in mathematics is not empty ritual. Prestige, venue, citation networks, and reputation are noisy but genuinely informative signals of prior achievement, reliability, and community knowledge about who has judgment where. Abolishing them would make abundance worse, since the scarce resource is trustworthy attention. The issue is calibration rather than legitimacy: those signals were calibrated when correctness, reactivability, and standing were correlated, and a signal reliable because it tracked three things does not announce when it begins tracking one. The requirement is **revalidation** — checking institutional signals against artifacts rather than letting them stand in for artifacts.

The two can come apart

Trace and aura, opposed in §3, dissociate in mathematical practice, and machine production makes the dissociation visible. A proof can carry every mark of its author's difficulty and be published by someone with no standing. A proof can carry every marker of institutional authority and have been sanded smooth until nothing can be learned from it.

The case neither Benjamin nor Tao quite states is now the default one. **Reproduction can destroy aura without preserving trace**, by two separate mechanisms: open formal verification weakens custodial authority, since a proof anyone can check against a published verifier needs no reputation behind it; and AI-polished exposition, independently, erases the marks through which the result might have become reactivatable. Run both together and formal checking becomes broadly available while human understanding stays systematically undersupplied.

The task is therefore not to sort aura into good and bad halves but to oppose cultic distance while preserving the traces that support a new act of understanding. Husserl's target is the draining of sense, which the trace resists; Benjamin's is the authority conferred by ritual administration, not the marks of the hand. Benjamin was writing about ritual and not tenure committees: cult value as the institutional correlate of aura is my construction, not his doctrine.

Hence: **destroy the cult, keep the trace**. Both halves need their narrow reading. “**Destroy the cult**” does not mean abolishing expertise, journals, referees, or reputation; it means removing non-epistemic dependence on proximity, affiliation, and privileged access while keeping expert judgment, provenance, and the recoverable trace. Expert judgment stays. What changes is that it operates on artifacts anyone can examine, through procedures anyone can contest.

“**Keep the trace**” does not mean publish every dead end. The principle is: *do not manufacture false fluency*.

Two qualifications. Reactivation can be distributed: for a large result no one person need hold all of it, provided every module has an accountable expert, the interfaces are understood, the architecture is explainable, and every essential step is understood by at least one accountable person. And the trace belongs in a companion artifact rather than the principal exposition — mathematics often advances by removing accidental difficulty and finding the right compression, and “keep the trace” must not license muddier papers.

That companion is the **reactivation packet**, and it holds what is needed to reconstruct why the successful path works:

- the governing proof idea, stated as an idea rather than a sequence of steps;
- the pivotal lemmas, marked as pivotal;
- the failed approaches that *materially explain* the successful one;
- the dependency map, including surprising dependencies;
- a recorded explanation by someone who can give one;
- the machine-verifiable artifact.

For many human-authored projects much of this exists already in working notes and is discarded at write-up, which is exactly when the trace is destroyed. For heavily automated work it does not exist at all unless the pipeline is built to retain it.

A machine can manufacture friction. Ask a model for a discovery narrative and it will supply plausible false starts, apologetic asides, and a well-shaped account of a struggle that never happened. So the packet must keep apart three things that look identical on the page: a **process-evidence trace** — versioned drafts, commits, proof-search and tactic logs, contemporaneous rather than reconstructed; a **retrospective rationale**, reconstructed after the fact, useful and not evidence; and a **difficulty map**, marking which lemmas or transitions were pivotal. The risk is not hypothetical, and the clearest documented instance comes from Tao’s own work. After deep-research tools proved effective at surfacing literature buried in obscure sources, he helped launch an effort to use them systematically to report the known literature on an open-problem site. The tools then treated those reports as an authoritative source, so that summarizing the searches interfered with any subsequent use of the same tools to find genuinely new literature — a circular citation loop, without anyone intending one.³⁵ Provenance can collapse through ordinary diligence.

All three help a reader. Only the first is contemporaneous, and even it is not self-authenticating: commits can be back-dated, logs curated, drafts regenerated. Absent cryptographic signing and trusted timestamps, the claim is comparative — a contemporaneous record is stronger provenance evidence than a retrospective narrative — not that discovery has been proved.

³⁵Klowden and Tao, “Mathematical methods and human thought,” §4.8, where the phenomenon is called “citogenesis,” after Randall Munroe.

Reactivation also needs a test, or it stays elastic enough to mean anything. A minimum public standard that makes the criterion contestable: can a competent reader state the governing idea without reproducing the derivation (**compression**), explain an arbitrarily chosen pivotal step (**interrogation**), apply the central idea to a nearby problem (**transfer**), diagnose a deliberately perturbed statement or proof (**repair**), and explain the result at two different levels of expertise (**translation**)? Passing raises the probability of reactivation; it does not guarantee the act, which remains something a reader does and not a property the artifact has. Unlike a recorded talk, these can be administered asynchronously by competent evaluators working from a public protocol.

Derrida in the margin

The word “trace” is borrowed and the loan should be acknowledged. Derrida’s first book was his translation of and introduction to *The Origin of Geometry*, and his subject there is reactivation itself.

His argument constrains mine. Writing is not a danger that befalls reactivation from outside; it is reactivation’s condition of possibility *and* the permanent possibility of its failure. Geometry survives its founder because marks are repeatable in new contexts — and the same iterability guarantees a mark can function with the founding act wholly absent. Derrida’s sharper formulation is that “nonpresence and otherness are internal to presence”: there was never a plenitude to fall away from.³⁶

So a machine proof that no one understands is not an aberration but the limit case of what inscription always was.

That forces a reformulation, and the reformulation is an improvement. Reactivation is neither recovery of an original act nor psychological re-enactment; it is a **new achievement, oriented by the deposit and never guaranteed by it**. Tao fighting through Bourgain recovered nothing of Bourgain’s consciousness. He produced his own understanding, occasioned by marks Bourgain left. The packet is not for restoring an origin but for improving the odds of a new act.

So the slogan takes a qualification: *destroy the cult, keep the trace — but do not mistake the trace for the origin*.

Three consequences

Make the demonstration the artifact rather than the audience. Tao’s criterion should be satisfiable by a recorded, publicly posted, expert-level exposition — checkable by anyone competent, requiring no room to be opened. This is exhibition value replacing cult value. It does not by itself preserve the trace: a talk demonstrates that someone can explain the result, and my own argument about manufactured friction shows that an explanation need not disclose where the work was actually difficult. The packet carries that; the talk carries the demonstration. Tao’s

³⁶The quoted phrase is from Jacques Derrida, “Signs and the Blink of an Eye,” in *The Phenomenology Reader*, 547–53, at 551, excerpted from *Speech and Phenomena*; it is not from the *Origin of Geometry* introduction or “Signature Event Context.” On iterability and the functioning of the mark in the absence of its producer, see Derrida, *Edmund Husserl’s “Origin of Geometry”: An Introduction*, and “Signature Event Context,” in *Limited Inc.* See also Mohanty, *Freiburg Years*, 425–27, esp. 427, on writing’s emancipation of ideal meaning from any actually present speaker.

appendix already lists a peer-reviewed video journal for research talks; he does not connect it to access, and the connection is there.³⁷

Mark epistemic state explicitly, and separately from provenance. Whether a result is proved, conjectured, or computationally explored is a fact about the result; who produced it is a fact about the author. Rendering both in the same typeface fuses the questions. In my own work this takes the form of a three-state badge carried in metadata, offered as an existence proof that Jaffe and Quinn’s 1993 recommendation is now mechanically cheap.

A zero-`sorry` discipline is not sufficient, and I have used it as shorthand for a verified state. An artifact can show no visible `sorry` and still depend transitively on `sorryAx`, on custom axioms, on native evaluation, or on a trusted base larger than a reader assumes.

The labels below are mine; their assurance ladder adapts the escalating validation model in Lean’s own guidance. **VERIFIED** should carry a reproducible build against a pinned toolchain, `#print axioms` on the principal results reporting only the standard axioms, the dependency closure and library commit hash, and disclosure of any custom axioms or native evaluation. **HIGH-ASSURANCE VERIFIED** adds rechecking with `lean4checker` and, at the top, sandboxed comparison against a separately trusted challenge statement using external checkers. Lean recommends that top tier not for ordinary work but for high-risk settings, and its own list of examples is worth quoting for how directly it lands here: proof marketplaces, high-reward proof competitions, and unaligned AI. The same guidance classifies un-reviewed AI-generated proofs within its *malicious* threat model — not as dishonest, but as the category that ordinary checks were never designed to catch.³⁸

Requiring the top tier of every result would be impractical; reserving it for load-bearing ones is not. Without any of these disclosures a badge does not assert nothing — the kernel checked something — but it leaves its trust assumptions unspecified, which is the same problem in a quieter form.

I can report what that looks like in practice, from my own corpus. Running a large generation-and-audit loop against a Lean development, I found twenty-eight files that satisfied the proxy exactly — they compiled, and contained no `sorry` — and contained no theorems at all. Under a harvester keyed to *compiles, no sorry*, empty files scored as passing proofs of famous open problems. Nothing was falsified and no proof was wrong; the metric was simply satisfiable without content. Goodhart’s law does not require an adversary, only a proxy and a loop.³⁹

Publish the criterion; open the measuring; announce both. This is not an outside demand. Among the values the Leiden Declaration lists as characteristic of mathematics is that arguments are transparent and independently verifiable, and that however long or difficult an argument may be, “no proprietary knowledge or equipment should be required to understand” it.⁴⁰

It names among the present threats the disadvantage borne by researchers without access to the relevant technologies or to the decisions about them, falling, it notes, disproportionately on students and early-career mathematicians.⁴¹ The Declaration also asks the community to

³⁷Tao, “Mathematics in the Age of AI,” appendix A, 12, listing *Mathematical Discourse*.

³⁸Lean Project, “Validating a Lean Proof,” *Lean Language Reference*, accessed 20 August 2026, <https://lean-lang.org/doc/reference/latest/ValidatingProofs/>.

³⁹An audit of the same corpus found no invalid proofs. The failure was entirely on the side of what the proxy did not require.

⁴⁰*Leiden Declaration*, 2, under “About our values.”

⁴¹*Leiden Declaration*, 3, under “Potential threats.”

make its standards and practices explicit and accessible, and to create pathways for meaningful participation.⁴² It carries the endorsement of the International Mathematical Union, and of Tao himself.⁴³ What follows is an attempt to say what those commitments require operationally, since a value declared but not implemented is indistinguishable from one not held.

SAIR’s Stage 2 judge is the model for the first: a deterministic verifier under an open license, with the evaluation set held back, makes the formal acceptance criterion reproducible at almost no cost to anti-Goodhart discipline. The second does not follow automatically and needs stating as its own requirement. Where a stage admits new entrants, the ability to be *measured* should open on the same terms as the ability to enter, since an institution that lets anyone check their work and scores only some has opened checking of the correctness criterion and left standing untouched. The third is the cheapest and the most easily skipped. Eligibility and tooling access should be versioned, dated policy, and when a restriction is lifted it should be announced on the channel where it was imposed. A correction that ships without notice reaches only those already watching — which is precisely the population the restriction was least binding on.

11. What this does not solve

The criterion defended here is reactivability, and enforcing it requires that someone competent actually reactivate. Under abundance, competent attention is the binding constraint — which means any rule of the form “someone must be able to explain this” is satisfied cheaply by whoever already has a queue of colleagues and expensively or not at all by whoever does not. “Any competent subject” states the right criterion; it does not tell an editor what to do on a Tuesday when volume exceeds the referee pool, and it does not prevent credentialed judgment from reappearing as the rationing mechanism.

I do not have a full answer. The partial mechanisms reduce load rather than eliminate it: recorded explanations checkable asynchronously rather than live by committee; semantic change logs that make most fidelity questions mechanical; adversarial statement review, narrower than proof refereeing; layered summaries that let readers allocate attention; and sampled expert auditing weighted by consequence rather than exhaustive review.

Two further limits belong on the record.

Both limits are named, independently, in Klowden and Tao. They describe a digital divide in which frontier tools are available chiefly to well-financed or well-connected research groups, and they observe that the published literature dwells on successful proofs while discarding the processes that led to them — which is exactly the material the reactivation packet asks authors to retain, and exactly why so little of it exists to train on.⁴⁴

Opening an artifact does not equalize the capacity to evaluate it. “Anyone can inspect the judge” is formal access. Compute, bandwidth, language, training, disability, time, and institutional support stay unequally distributed, and a public repository redistributes none of them. Publishing the criterion is necessary and nowhere close to sufficient.

The labor this paper asks for is unpaid. Exposition, refereeing, transcription, annotation, adversarial statement review, canonicalization — every mechanism proposed here adds work to

⁴²Leiden Declaration, 6, “Welcome new contributors.”

⁴³Endorsements are listed at <https://leidendeclaration.ai/>; for the IMU’s, see also *IMU News* 137 (May 2026).

⁴⁴Klowden and Tao, “Mathematical methods and human thought,” §5.3 and n. 12.

roles the profession already treats as less prestigious than proving theorems, and adds it under abundance, when the volume is climbing. Stein transcribed Husserl's shorthand and got no chair for it. A proposal that raises demand for transmission labor without addressing how it is compensated and credited is asking the same people to absorb the cost again. I do not solve that here, and it should not be filed under attention scarcity, which makes it sound like a scheduling problem.

That is a research program, not a solution. The honest position is that the diagnosis here runs ahead of the institutional design, and the design is where the work now is.

12. Coda

Among the most exacting accounts we possess of what it means for a science to understand itself was produced by a man whom his own university was erasing.

The record is short and I set it down once. Husserl retired from Freiburg in 1928; his former assistant Martin Heidegger, whose career he had supported, succeeded him in the chair. The University withdrew Husserl's teaching license on 6 April 1933. Heidegger became rector weeks afterward and joined the Party. He did not order the withdrawal — and in the University's own present-day words, he placed himself at the service of the new rulers and did nothing to protect his teacher. From 1936 Husserl's name was absent from the university directories. His death in April 1938 went officially unacknowledged by the institution whose phenomenological reputation he had helped establish. The *Crisis* appeared abroad because it could not appear at home.

Before Heidegger succeeded Husserl in the chair, Edith Stein had served as Husserl's first paid assistant, transcribing and substantially editing the research manuscripts that became *Ideas II* and the lectures on internal time-consciousness — which is to say she performed, literally, the labor of making his thought legible to anyone else. She never held a chair. She was murdered at Auschwitz-Birkenau in 1942.⁴⁵

Heidegger inherited the chair and the standing attached to it. Stein performed the labor of transmission and inherited none of that standing.

The transmission of Husserl's corpus did not follow the chair. It depended on a chain of labor the chair neither guaranteed nor adequately recognized: Stein's transcription and editorial reconstruction, Fink's collaboration on the late work, the Euckens' personal support, and Van Breda's rescue of the archive — some forty thousand pages of shorthand carried out of Germany to Leuven. Different tasks, different decades, one thing in common.

The disanalogy must be explicit. Nazi persecution and murder are not analogies for a competition rule, and I am not trading on them. The historical case establishes one proposition only: **standing, capacity, and preservation can come apart.** The contemporary design requirement follows from that separation, not from the magnitude of the history.

⁴⁵Details from the University of Freiburg's own account of Husserl (see References) and, for Stein, Szanto and Moran. Stein's 1919 habilitation attempt failed amid the effective exclusion of women from the professoriate; she was dismissed from her post at Münster in 1933 for her Jewish ancestry, entered the Carmel at Cologne, moved to Echt in the Netherlands, and was arrested in August 1942.

Exclusion and incapacity are different things. An institution that uses access as a proxy for capacity cannot reliably distinguish them. The capacity for reactivation must therefore remain distinct from access to the settings in which it is demonstrated.

Tao is right that mathematics is entering a period comparable to 1900–1930, and right that codification will follow. But the first crisis concerned the foundations of **truth**, and it was institutionally stabilized. The one he describes concerns the conditions of **understanding**. It was named in the same decade by someone who diagnosed the structure of such a loss; a second author that year named what goes wrong when we defend understanding by defending the position of its keepers; and a third specified the machine and then marked its limit.

We are not starting from zero. A judge can establish that an artifact satisfies a published formal criterion relative to a stated trusted base. What remains is the human work of making the sense present again as understanding — and of ensuring the making is open to whoever can do it.

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Tool and computational resource disclosure. This paper was drafted on 19–20 August 2026, over approximately two days, with the assistance of several artificial intelligence systems: Anthropic’s Claude (Fable 5 and Opus models), OpenAI’s ChatGPT (5.6 Sol), xAI’s Grok (4.6), and an orchestration layer of the author’s own construction. Their contributions were structural criticism, adversarial review under assigned hostile and disciplinary perspectives, drafting alternatives, bibliographic discovery, and verification of locators against source files supplied by the author. Successive drafts were subjected to simulated review from formal-methods, critical-theory, phenomenological, and research-mathematics perspectives, and the manuscript was revised in response. No system is an author, and no system bears responsibility for anything asserted here.

The argument, the six-relation framework, the underdetermination argument for statement fidelity, the reactivation packet, the reading of Benjamin’s trace/aura opposition against Tao’s natural friction, and all interpretive claims are the author’s, as is final editorial responsibility.

The compressed timeline is disclosed deliberately, since a paper arguing that provenance should be inspectable rather than inferred ought not to conceal its own. Nothing here is offered on the strength of how long it took. Every load-bearing claim is attached to a locator a reader can check independently, which is the standard the paper asks of others.

On sources: the author has personally consulted, in the editions cited, Tao’s essay; Turing’s two papers in *The Essential Turing*; Jaffe and Quinn; Thurston; the Leiden Declaration; Benjamin’s essay in *Illuminations*; Mohanty’s *Freiburg Years*; and *The Phenomenology Reader*. Pagination was not independently verified in Husserl’s *Crisis* (Carr), the *Logical Investigations* (Findlay/Moran 2001), Derrida’s *Introduction to the Origin of Geometry* and “Signature Event Context,” Benjamin’s *Arcades Project*, or Gödel’s *Collected Works* vol. III. Claims resting on those works are therefore cited by stable section or convolute rather than by page — Husserl’s and Turing’s section divisions and Benjamin’s convolute numbers hold across editions — and are cross-checked against the secondary scholarship identified in the notes.

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